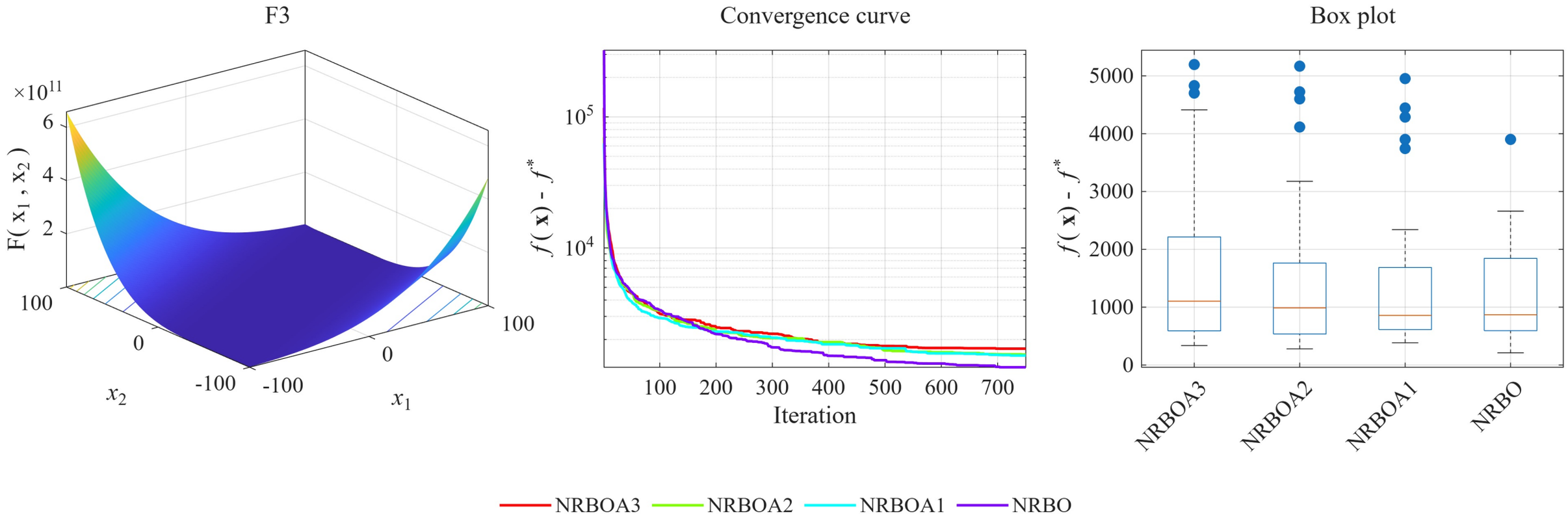

F3
×10^11
6
4
2
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^5
10^4
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
5000
4000
3000
2000
1000
0
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO